2. ESTIMATION OF PROTEIN (BIURET METHOD)

Aim

To estimate the amount of protein present in the given test sample.

Principle

The-CO-NH- group of protein forms a purple coloured complex with copper ion in alkaline medium. The colour intensity is measured at 540nm. Since all proteins contain peptide bond, this method is fairly specific and there is little interference with other compounds.

Reagents required

Stock Biuret reagent

Dissolve 45g of Rochelle's Salt (Sodium potassium tartarate) in about 400ml of 0.2N sodium hydroxide and add 15g of copper sulphate. Stir it continuously until the solution is complete. Add 5g of potassium iodide and make upto 1 litre with 0.2N sodium hydroxide.

Biuret solution for use

Dilute 200ml of stock biuret reagent to 1 litre with 0.2N sodium hydroxide containing 5gm of potassium iodide.

Stock Standard Solution

1g of protein(egg albumin) is weighed and made upto 100ml with distilled water.

Concentration = 10mg/ml

Working standard solution

10ml of the stock is diluted to 100ml using distilled water.

Concentration = 1mg/ml

Procedure

Estimation of protein

0.5-2.5ml of standard protein solution is pipette out into five different test tubes (S1-S5). The concentrations of protein in the tubes are 0.5-2.5mg. 0.5 and 1.0ml of test solution is taken in two test tubes labelled as T_1 and T_2. The volume in all the tubes are made upto 5ml using distilled water. A blank is also prepared simultaneously by adding 5ml of distilled water. Then 3ml of biuret reagent is added to all the test tubes including blank. The tubes are mixed well. The tubes are then maintained at room temperature for 10 minutes. The optical density is measured at 540nm.

From the values obtained, a standard graph is drawn using concentration of protein in X-axis and optical density in the Y-axis. From the graph the amount of protein present in the given test solution is calculated.

Practical_Bio-Chem-EM.indd 269 4/8/2020 3:28:52 PM